Final Report: CAD Project
Limbo

Presentation: https://youtu.be/_5wkbBfN5GE

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Abstract

Our report displays our process for designing engineering parts using CAD software. In this project, we went through the steps to sketch, design, and model a robotic arm according to the guidelines. The goal of this project was to develop a practical and manufacturable robot arm while considering the range of motion, part interaction, stability, and more. By researching the history of robotic arms, we were able to learn from and build on previous designs and apply them to our own robot arm design. In the report, we have included sketches of design concepts that initially helped us get a general idea of what we wanted our arm to look like and the functions we wanted in our manipulator. These concepts were taken into careful consideration and refined to meet the constraints of manufacturing and the rubric. After designing the entire arm, we made theoretical calculations to determine the dimensions, range, and overall capabilities of our robotic arm. With our final design draft and calculations completed, we modeled and made drawings of our parts using CAD software and assembled our robotic arm. The assembly process allowed us to identify nuances and problems with our original design, showing us exactly what issues needed to be dealt with. The conclusion reflects our results, explaining the positives and negatives of our design and design process.

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1. Introduction and General Background

Robotic manipulation is a foundational concept in engineering, robotics, and automation, with applications in many industries like aerospace, manufacturing, and medical devices. The most common relationship about robotic arms that people know about seems to be manufacturing robots that assist in factories and assembly lines, whether that be for small products or even large ones like cars. While the large industrial robot is very impressive in an engineering sense, they are also expensive and generally inaccessible to students seeking to learn about mechanical design and the foundational concepts of robotics. On the other hand, smaller and low-cost robot arms may be of value for students to engage with the different considerations that come with building a robot with several degrees of freedom. Some of these considerations include torque and load analysis, structural design, gear design, and joints.

Our team's objective is to develop a 6 degree of freedom robot arm, consisting of a rotating base, a primary link, an elbow link, and a claw-like gripper at the very end, with the goal of picking up and repositioning small objects. While this project does not necessarily address an industrial or commercial need, it provides an opportunity to apply mechanical design concepts and processes like torque analysis, gear ratio selection, and structural design in a more achievable and hands-on way. Using mainly Polyethylene terephthalate (PET), we hope to make this project accessible by having 3-D printed parts and metal parts that you can either manufacture yourself or look for a similar alternative online. We are excited to showcase the discoveries we made, displaying what worked and what might have worked, as well as improvements for future projects.

2. Design Goals and Objectives

The design goals and objectives for this project are outlined below:

The device must:

(1) Have six degrees of freedom

- Translation - x, y, and z axes
- Rotation - Roll, Pitch, Yaw
- At least one of the degrees of freedom needs to be rotational

(2) The arm motion should be controlled by an electric motor

(3) The design must be an assembly of at least 5 sub-assemblies

(4) Dimensions must be stated both in metric and imperial units

(5) Theoretical calculations must be done with regard to the design's functionality and tasks

(6) Each sub-assembly must be concluded from these calculations

3. Prior Work (State-of-the-art) or Scientific History

The first usage of the word "robot" originated from a Czech play in 1921 and was eventually coined by a Russian-American writer. In the early 1950s, the first industrial robotic arm, the **Unimate**, was invented and patented by an American engineer, George Devol. In his patent application, Devol mentions that his "general-purpose machine" had universal application to manufacturing. Founded by Devol and Engelberger, Unimation made the world's first robot and was put to the test in a General Motors assembly line as a materials-handling machine. After years of testing, Unimate started full-scale production in Connecticut in 1966.

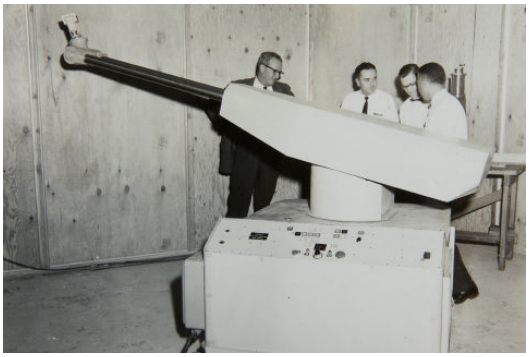
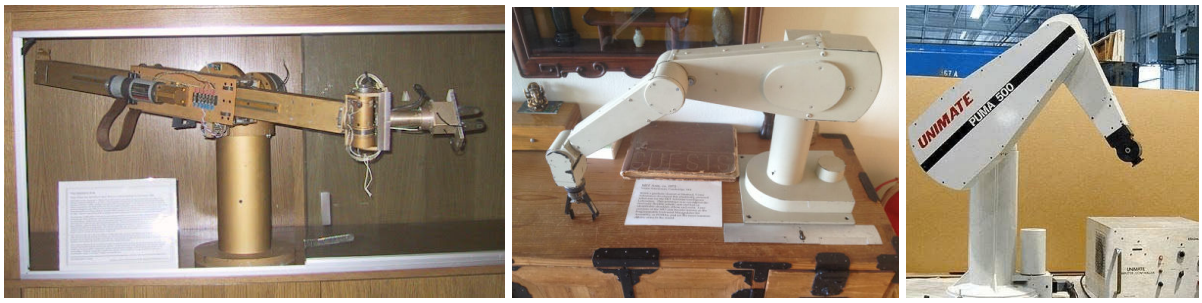


Figure 1: The Unimate

In the years following Unimate, a few other arms were developed. For example, there was the Rancho arm, a six-jointed arm to assist the handicapped, the tentacle arm designed by Marvin Minsky, and the spray painting arm by ABB. The most notable after Unimate was the **Stanford Arm**, designed by Victor Scheinman. Making its debut in 1969, the Stanford Arm led to mainstream commercial production of robot arms. It was able to evolve at an alarming rate and was the first electrically powered, computer-controlled robotic arm. This arm was eventually merged with the **MIT arm** in the 1970s, founding the company Vicarm, Inc. for production. Scheinman also contributed to the development of the **PUMA** (programmable universal manipulator assembly) for Unimate.



Figures 2, 3, 4 from left to right: Stanford Arm, MIT Arm, PUMA

The robot industry officially takes off in full force in the 1970s. However, large investments from automotive companies in the 1980s cost automotive companies such as General Motors a lot in the 1980s as mishaps began piling up. The premature introduction of robots cost companies a lot of money and allowed for Japanese and European companies to take over and overshadow the American market, and robot sales across the globe increased again in the 2010s.

4. Concept Development

4.1 Design Concept 1

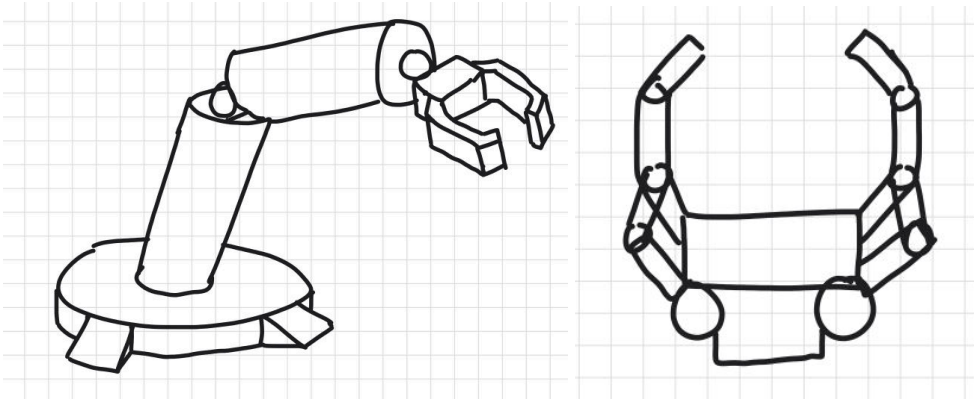


Figure 5: Schematic of the initial design

4.2 Design Concept 2

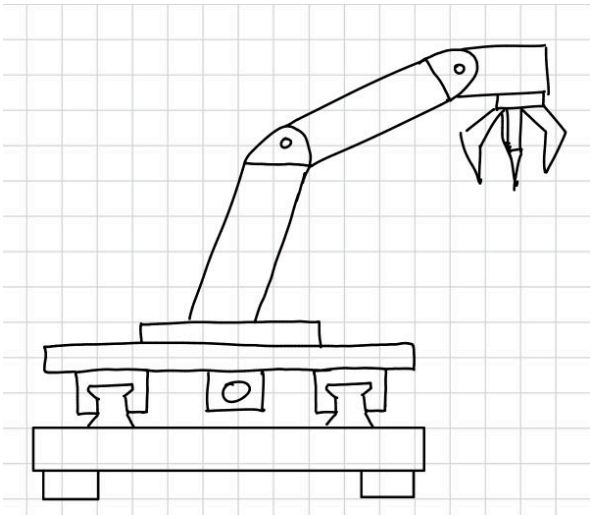


Figure 6: Schematic of Second Design

Our second design saw changes with the base and claw. The base would be secured onto linear guides with a mounting carriage. The linear guides would allow for a wider range of horizontal movement in a single axis. A claw grappler with 4 prongs would enable the arm to have a stronger grip of objects moving up and down the z-axis. For this design, we would have needed linear guides, a ball screw, and either another motor or ball bearings for the 4-pronged claw.

4.3 Design Choice

Our group's final design decision was to have a 2-jawed parallel gripper, a rotating base and elbow, and hinging limbs. This design allowed us to create an arm with 6 degrees of freedom. The rotating base provides a

rotational degree of freedom, while the elbow joints allow the arm to extend, retract, and move vertically. The 2-jawed end effector was selected as a simple yet effective design for grasping materials. Together, these features blend in a way that balances mobility, functionality, and mechanical simplicity while meeting the requirements.

5. Theoretical Calculations

Using solidworks, the total mass of the robot arm, which was mostly made of Polyethylene terephthalate (PET), comes out to be about 4159g, or 4.159 kg. Using 6061 Aluminum, the total mass is around 5567g, or 5.567kg.

For the gear ratio of the planetary and stationary spurs between the lower limb and elbow link, we considered the need for speed of the arm as well as the torque requirement. Opting for a 2:1 gear ratio, where 2 represents the large stationary gear and 1 represents the smaller motor-driven gear, we tried to get as close to the ratio as possible. Using the distance With a center distance of 62mm and standard module of 2mm, we used the equations:

$$C = (N1 + N2) \times (m/2)$$

Where C is center distance, N1 and N2 are the number of teeth for the respective gears, and m is the standard module size. Combined with the standard gear ratio equation and requiring integers for number of teeth:

$$GR = 2:1 = N1/N2$$

Adjusting closely to the ratio, the number of teeth acquired is N1 = 42 teeth and N2 = 20 teeth, which comes out to a ratio of 2.1:1.

With the gear ratio found, the maximum load lifted by the elbow link can be found, with the help of some motor specifications and calculations. Given the estimated rated torque 0.5 N-m, which is the sustainable output torque the motor can deliver indefinitely, we can find the output torque of the motor-gear system by multiplying this value with the gear ratio. (0.5 N-m = 5.0986 kg-cm)

$$5.0986 \text{ kg-cm} \times 2.1 = 10.707 \text{ kg-cm}$$

Taking into account the weight of the elbow link and claw themselves, that number reduces to 1.74 kg-cm after torque calculations from center of mass of different subassemblies. At the distance where the object is held in the claw, the greatest mass that can be carried is about 46.78 g, which is not very much. If the entire assembly were in full horizontal extension, the motor would not be able to hold.

Overall, after looking at the calculations, revisions on the design would greatly improve the maximum load lifted, especially for the lengths of the links. Furthermore, a stronger motor or use of multiple motors for the base joint would also help resolve the issue that the arm faces at full horizontal extension.

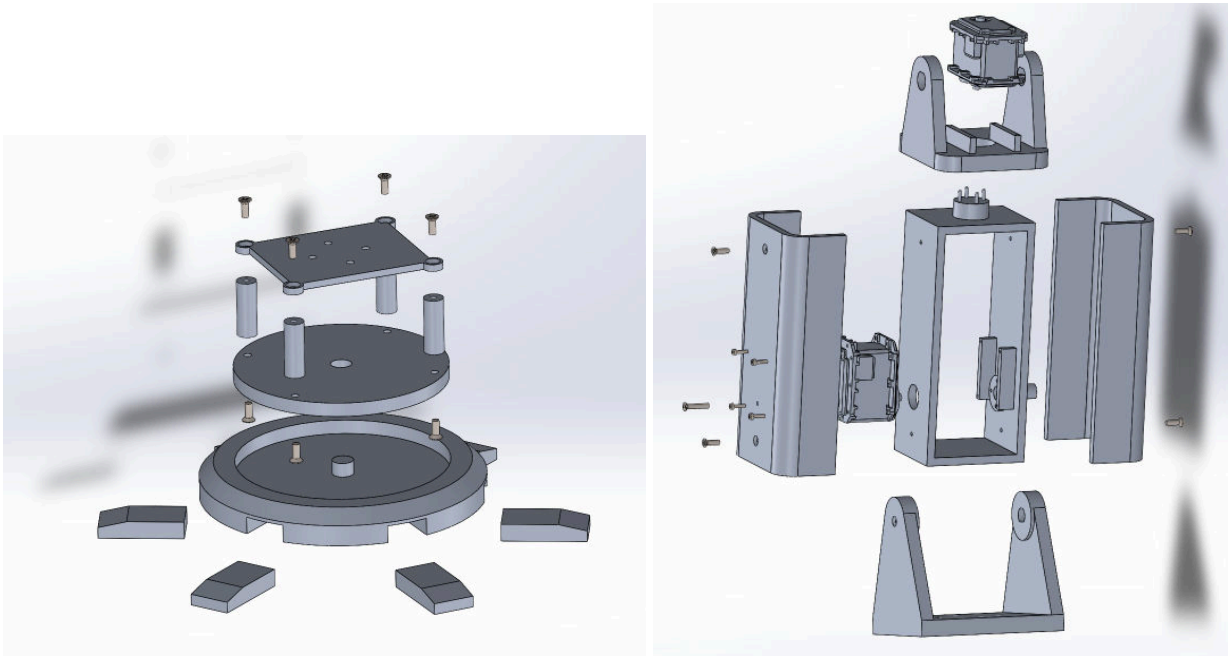
6. Design Specifications

Total Unique Parts (not including toolbox)	26
Total Components	116
Subassemblies	4
Weight (6061 Aluminum)	5567.30g
Max Height	750mm
Degrees of Freedom	6
Motors	3
Power Requirements	12V/1.2A DC per motor

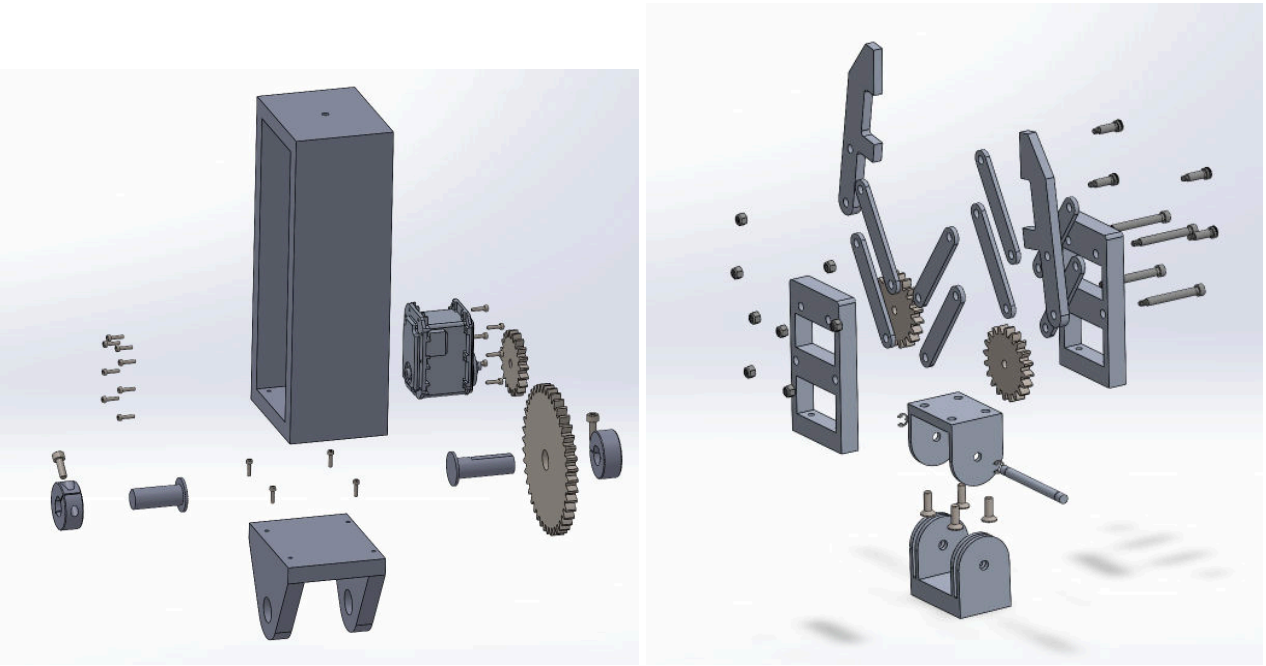
Table 1: Design Specifications

6.1 Final Design Description

In this section, we discuss how the robot arm functions across the different subassemblies and the role their respective parts play. Additionally, we will be going over how the arm is powered.



Figures 7 & 8: The Base and Limb 1 Subassemblies Exploded Views



Figures 9 & 10: Limb 2 and Claw Subassemblies Exploded Views

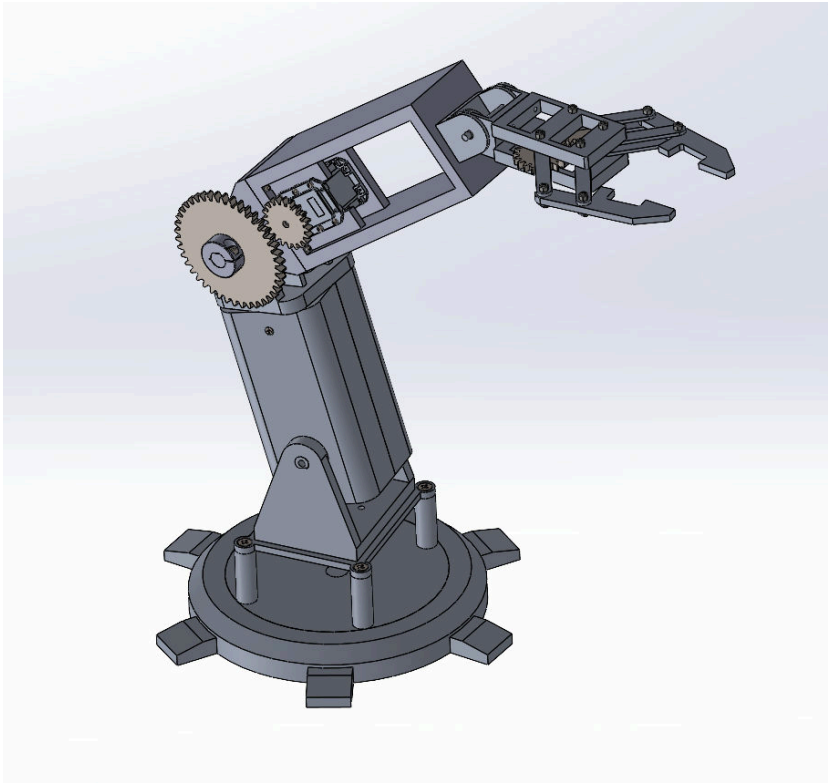


Figure 11: Solid model of the final assembled device

6.2 Device Parts Design

Base

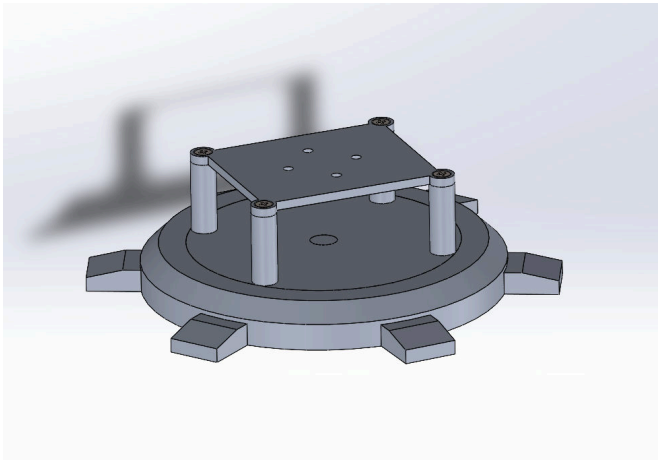


Figure 12: Base Assembly Isometric View

Base Parts	
1	Base Disk
2	Base Feet
3	Base Arm Platform
4	Base Mounting Peg
5	Base Mounting Plate

Table 2: List of Parts in the Base Assembly

1&2 - Base Disk & Feet

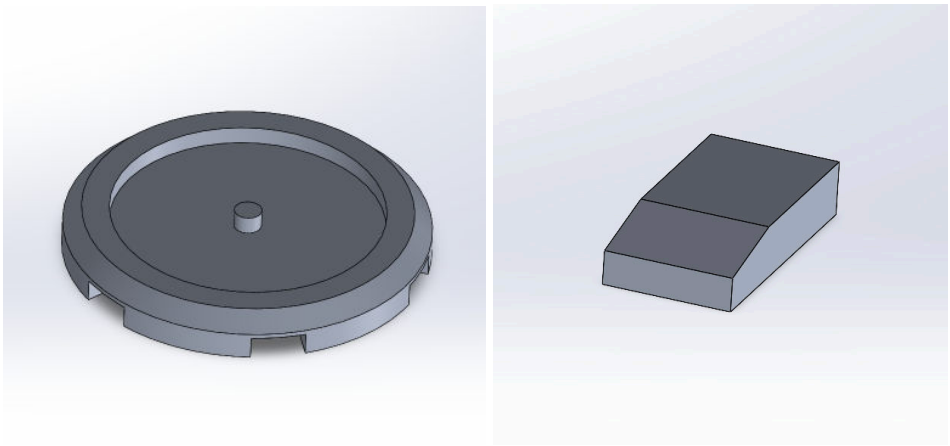


Figure 13 & 14: Base Disk & Base Feet in Isometric View

The Base Disk is designed to be the largest and heaviest part of this project, allowing the arm to stay planted on the surface that it sits on. It is relatively simple to manufacture. On the underside there are six rectangular slots for the base feet to fit in. The base feet are also very simple and the tolerances are designed for it to fit under the Base Disk via friction. It extends the courtyard area of the base, to prevent the arm from toppling over.

3, 4, & 5 - Base Arm Platform, Mounting Peg, and Plate

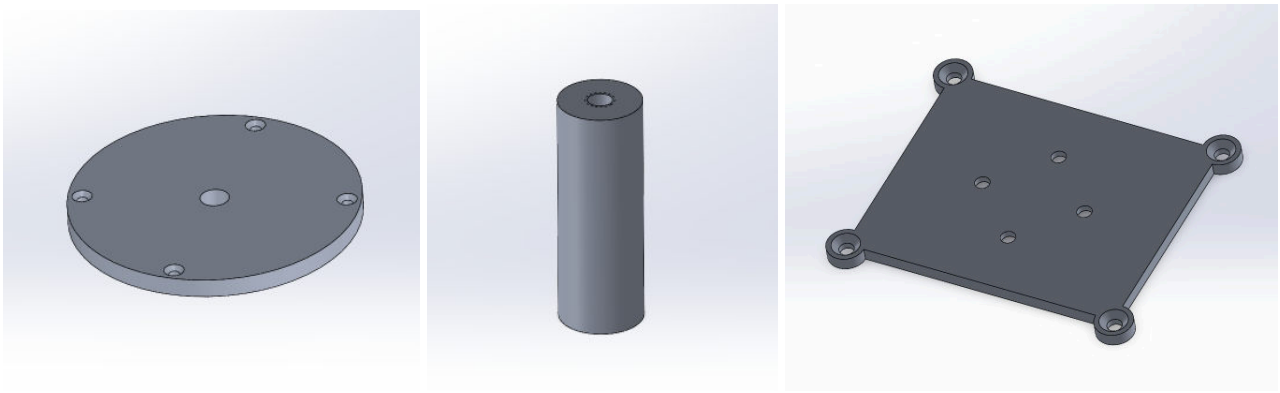


Figure 15, 16, 17: Base Arm Platform, Mounting Peg, and Plate in Isometric View

These three parts fit together with 8x M5 x 0.8 x 13 Cross Recessed Screws and serve as the platform which the Lower Limb connects to the Base. The Mounting Pegs situated on the corners of the Plate allow easier assembly, and also lifts the arm up for some extra reach. The Base Arm Platform has a hole that allows it to rotate on the Base Disk.

Lower Limb

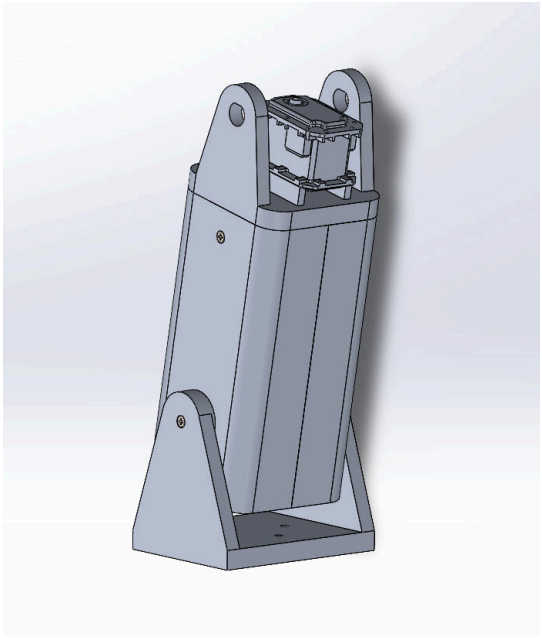


Figure 18: Lower Limb Assembly Isometric View

Lower Limb Parts	
1	Lower Limb Structure
2	Lower Limb Shell
3	Lower Limb Mount
4	Lower Limb Motor Axle
5	Lower to Second Limb Mount

Table 3: List of Parts in Lower Limb Assembly

1 - Lower Limb Structure

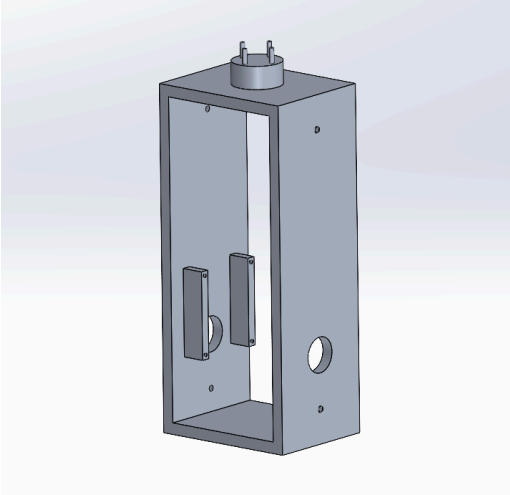


Figure 19: Lower Limb Structure

The Lower Limb Structure is the skeleton that holds the Lower Limb Shell pieces together. It also houses a motor mounted inside that controls the vertical angular movement of the limb relative to the base. The motor is mounted to the inside of the Lower Limb Structure using 4x 2 x 0.4 x 8 Hex Screws. At the top fo the Lower Limb Structure, it is shaped to accept a motor to fit onto the structure to allow side-to-side horizontal motion between the Lower Limb and the Elbow Link.

2 - Lower Limb Shell

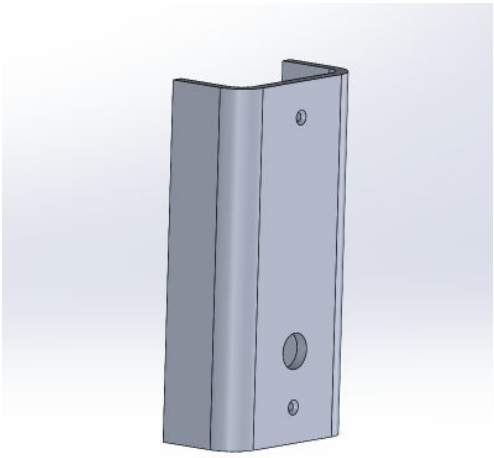
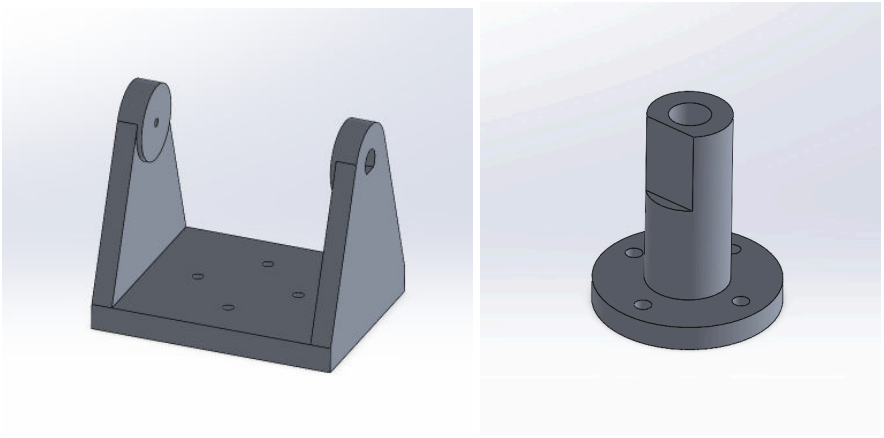


Figure 20: Lower Limb Shell (Default Config) Isometric View

The Lower Limb Shell consists of two different pieces stored in the configurations “Default” and “Right Side”. The Default configuration has a large hole for the Lower Limb Motor Axle to fit into for the motor to be able to drive vertical rotational movement. The two Lower Limb Shell each cover half of the Lower Limb Structure, and are mounted via 4x M3 x 0.5 x 10 Recessed Screws. The screws are recessed as to not interfere with the mounting system to the base.

3 & 4 - Lower Limb Motor Axle and Mount



Figures 21, 22: Lower Limb Mount (Left) & Lower Limb Motor Axle (Right)

The Lower Limb Mount and Motor Axle help connect the Lower Limb to the Base, while also giving the motor situated inside the Lower Limb Structure the ability to move the Lower Limb vertically up or down. The Lower Limb Motor Axle connects directly to the motor and fits into the right side of the Lower Limb Mount as shown in the figure. The Lower Limb Mount connects its other side to the Lower Limb Shell and Structure via a M3 x 0.5 x 16 Recessed Screw. The hole on the Lower Limb Mount and the threaded hole in the structure are designed such that minimal friction occurs, alleviating stress on the motor. It is recommended to lubricate the Lower Limb Mount's holes during assembly.

5 - Lower to Second Limb Mount

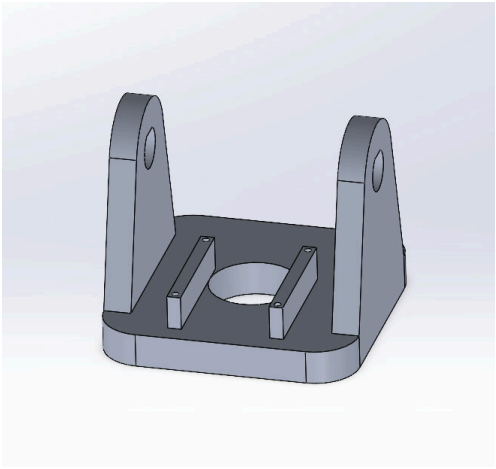


Figure 23: Lower to Second Limb Mount Isometric View

The Lower to Second Limb Mount has a hole in the center for it to sit on top of the Lower Limb Structure. A motor is mounted on top with 4x 2 x 0.4 x 8 Hex Screws. This motor allows side to side movement for the Elbow Link. The bottom of the Elbow Link connects directly to this part.

Elbow Link

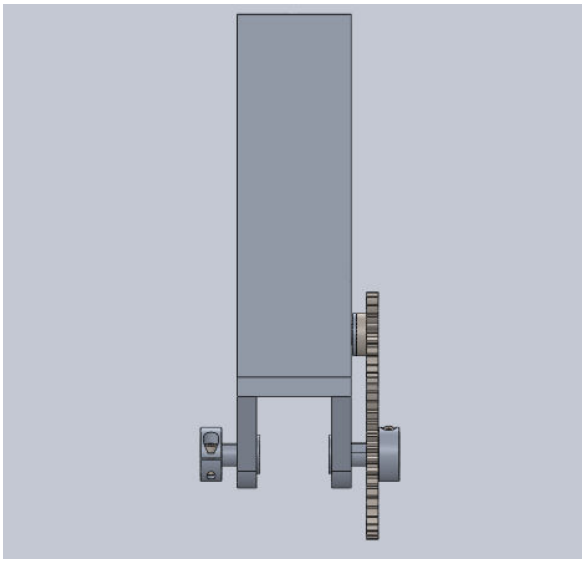


Figure 24: Elbow Link Assembly Front View

Elbow Link Parts	
1	Elbow Link Structure
2	Elbow Bottom Joint
3	Elbow Joint Shaft
4	Elbow Non-gear Joint Shaft
5	Elbow Stationary Spur Gear
6	Elbow Planetary Spur Gear
7	Elbow D-Shaft Collar

Table 4: List of Parts in Elbow Link Assembly

1 - Elbow Link Structure

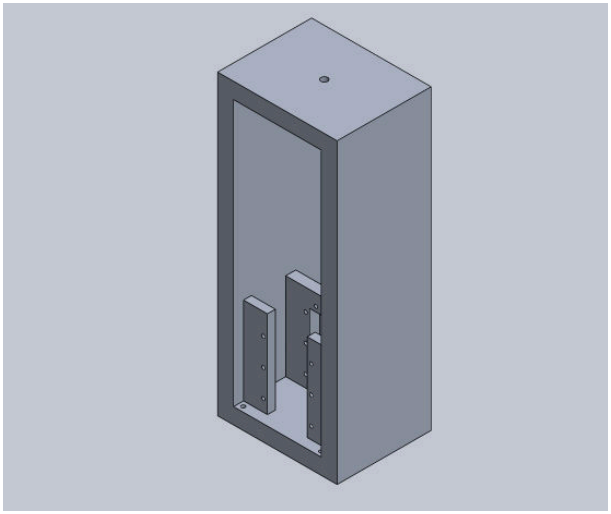


Figure 25: Elbow Link Structure Isometric View

The Elbow Link Structure is the main body of the assembly where all the other parts attach to. It is an extension from the first link that allows it to bend in an additional degree of freedom and reach farther objects. The structure is a hollow rectangular shell, which reduces its weight in comparison to the first link while maintaining the strength needed to lift loads. The purpose of this design is to keep the geometry simple and lessen the load on the first link and motors by reducing the material used. Besides the main geometry of the structure, there is also the bracket-like pieces sitting at the bottom of the part, which is used to screw the motor onto the part and keep it tight in place. Additionally, there are tapped holes going through the bottom of the structure to allow it to be mounted on another joint piece.

2 - Elbow Bottom Joint

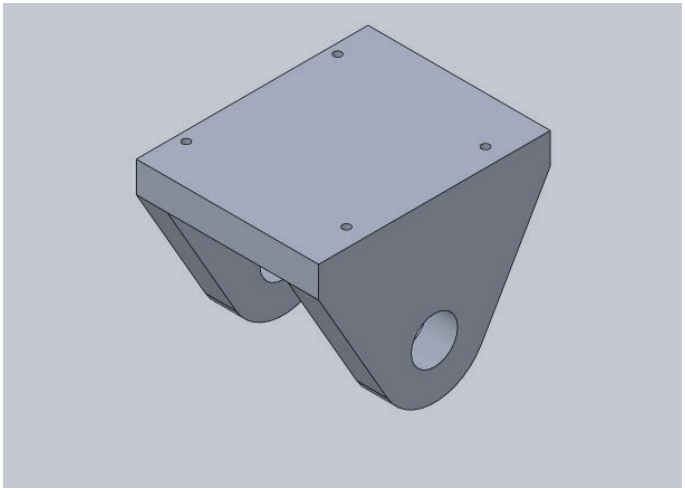


Figure 26: Elbow Bottom Joint Isometric View

The Elbow Bottom Joint is the connecting piece between the Elbow Link Structure and the main link's joint piece. The geometry of the part's top is designed to match the bottom of the Elbow Link Structure, matching the four tapped holes for assembly. The two shaft holes allow for one degree of freedom, which would be pitch/roll in this case.

3&4 - Elbow Joint Shaft (Gear and Non-gear)

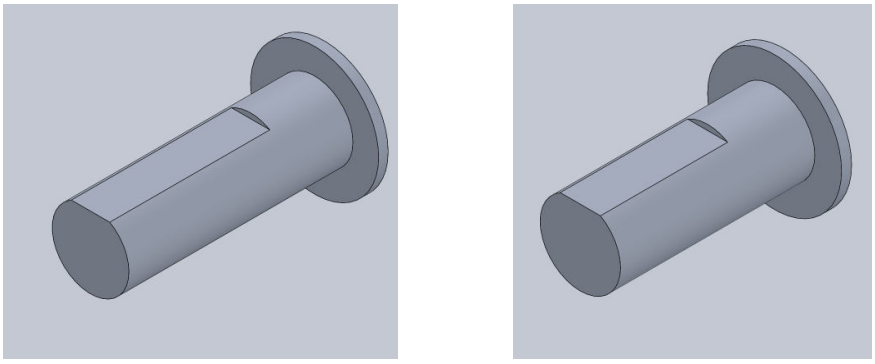


Figure 27 & 28: Elbow Joint Shaft Gear (Left) and Non-gear (Right) Isometric Views

Both of the elbow joint shafts serve the same purpose of connecting the joints between the elbow link and the main link and allowing them to rotate relative to each other. The grooves on both designs make them d-shafts, which is a type of shaft that promotes greater bearing force between the shaft and the contact component because of its geometry. Compared to a completely circular shaft, there is almost no slipping since it is incapable of rotating without rotating the component. Noticeably, there is a part of the shaft that does not have the d groove geometry, which gives freedom of rotation (as long as friction is minimal with the help of bearings). The idea behind the design is that the shaft will be fixed to the Lower to Second Limb mount, while the Elbow Bottom Joint will be able to rotate freely relative to the Lower Limb. One difference to note, however, is that the Elbow Joint Shaft is slightly longer than its twin to accommodate for the gear it will have.

5 - Elbow Stationary Spur Gear

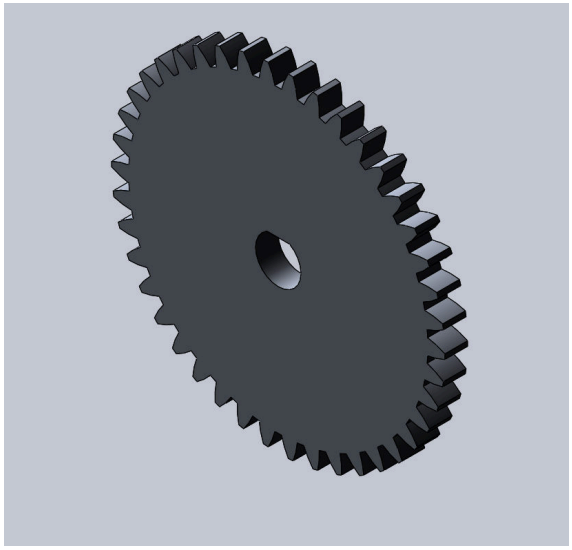


Figure 29: Elbow Stationary Spur Gear Isometric View

To keep the gear interactions fairly simple while leaning towards a slight advantage in lifting heavier items, we decided to get close to a 2:1 ratio. To get this ratio, we give the motor the smaller gear, meaning this stationary

gear must be larger and have more teeth, 42 to be exact. Our robot arm does not need to especially move fast, thus it would be advantageous to trade speed for load lifting capability. To prevent the gear from moving, we employed the d-shaft design again, making the bore hole match the joint shafts as discussed earlier.

6 - Elbow Planetary Spur Gear



Figure 30: Elbow Planetary Spur Gear Isometric View

As the stationary gear's counterpart, the planetary spur gear was designed to have 20 teeth, giving us a 2.1:1 ratio. In contrast to the stationary gear, this design has a hub that extends from the main body to connect to the motor, which is not exactly aligned with the stationary gear on its own. With a total of eight screw clearance holes, this part is meant to bolt tightly to the motor, giving the motor the ability to rotate the elbow link about the lower limb as a result of the gear's "planetary" revolution of the stationary gear.

7 - Elbow D-Shaft Collar

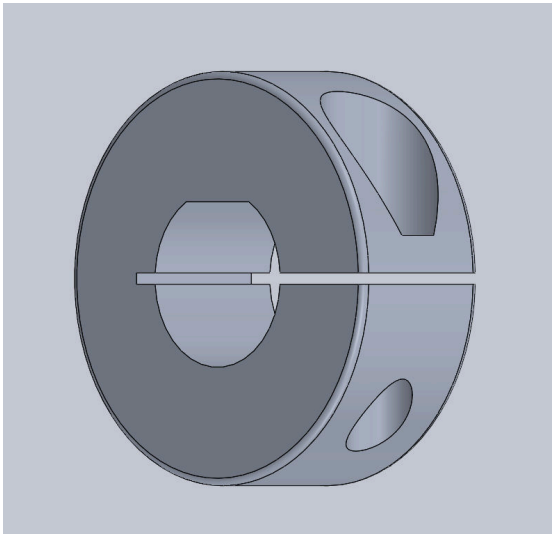


Figure 31: Elbow D-Shaft Collar View

Accompanying the previous d-shaft designs, this part helps to clamp the end of the Elbow joint shafts, preventing the stationary gear and respective joint and mount parts from sliding off or around the shaft. The

counterbore hole helps the collar clamp when a screw is tightened hard enough. To deform and tighten more easily, a groove and a gap has been added.

Claw

Claw Parts	
1	Claw Base
2	Claw Connecting Link
3	Claw Grippers
4	Claw Gears
5	Claw Joint Mount
6	Limb to Claw Joint
7	Claw Joint Connecting Rod

Table 5: Parts in Claw Assembly

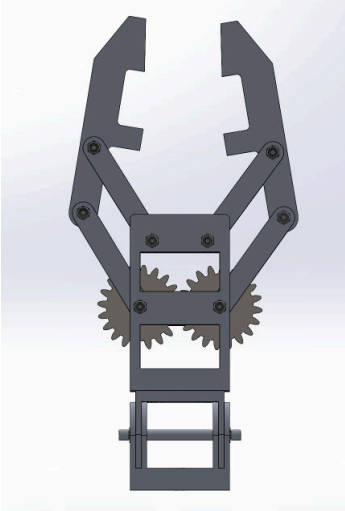


Figure 32: Claw Parts Table and Front View of Claw

The claw's design allows the arm to grip objects with two jaws that open and close in sync. Its front grippers move in unison, creating an opening and closing motion as a result of the two connecting gears attached to the base. These gears have a 1:1 ratio and are bound to the grippers using 8 connecting links, with four on each side. All of these parts are connected to the base using 30 mm and 10 mm long shoulder screws and nylon locknuts. This ensures that the parts can still rotate freely while securing them in a snug position.

The claw is attached to the rest of the arm using a joint consisting of two sections and a rod. The joint attaches to the base of the claw using four M4 socket countersink screws. To allow for free hinging movement, the joint is connected to the base using countersink screws. The joint is fully assembled with a rod joining the two halves and is attached to the limb using a washer and M6 screw. This means that the claw will not have a rotational degree of freedom, but will still hinge without the need for another motor.

6.3 Powering the Device

The arm is powered by a Dynamixel MX-28AR, which has an estimated rated torque of 0.5 N-m, about 20% of the stall torque 2.5 N-m.

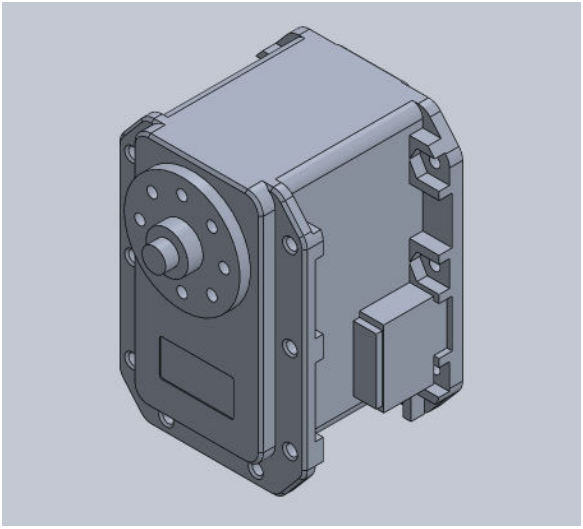


Figure 33: Dynamixel Motor Isometric View

7. Manufacturing and Fabrication -

Describe the manufacturing process for each parts, the material used, number of parts made, post-fabrication detailing, and any possible cost each process may undertake.
All parts designed for this project are best made through FDM 3d printing, using rigid filaments such as PLA or PETG. Screws and nuts are standard and should be sourced directly. To improve the durability of the arm, it is important to consider the rotation of the parts on the print bed so that the part can handle different forces in use. An example of the optimal orientation when printing is this lower_limb_shell piece shown below. This is the cheapest and easiest manufacturing option, with material costs not exceeding $15.

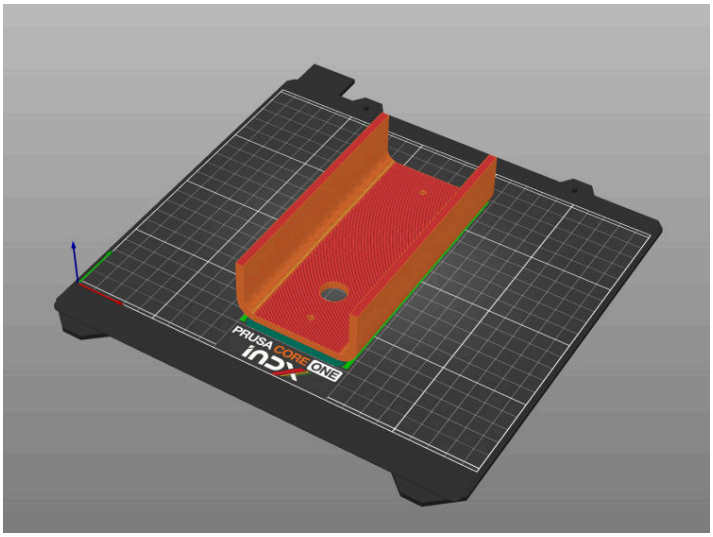


Figure 34: Lower_Limb_Shell positioned on a 3d printer bed

Besides 3d printing, using a CNC milling machine or a Lathe is useful to create this arm out of 6061 aluminum, which is what we used for each part's property calculations. 6061 Aluminum is easy to cut, and is significantly more durable than plastics from 3d printing. By creating the parts out of aluminum, the parts will need to have

their threads tapped manually. However, the total costs are estimated to be $400-500 total for all parts, not including labor.

The hole and shafts for the arm's joints are often the basic size, and depending on the manufacturing process, the holes and shaft's dimensions need to be adjusted as follows:

Manufacturing Process	Hole Tolerance	Shaft Tolerance
3D Printing	+0.15mm	-0.15mm
CNC Milling	+0.05	-0.05

Table 6: Assembly Tolerances for Different Manufacturing Methods

Upon finishing the shape of each part, no post-fabrication processes are needed other than the optional dielectric grease in the joint parts.

8. Conclusions

The completed assembly of our robotic manipulator consists of 26 unique parts. Designing the robot arm involved research, concept drafting, solid modeling, and creating part and assembly drawings. All parts designed are 3D printed using PET except for the gears and motors.

To begin our project, we researched the history of robotic manipulators, exposing ourselves to previous design flaws and strengths. We learned about the Unimate, the evolutions following (the Stanford arm and PUMA), and the European and Japanese industries taking over.

Our group made 2 design concepts after completing our research phase. We compromised on a design that applies major features from only 1 of the designs. The final design implements a circular base, lower and upper limbs, rotating and hinging joints, and a 2-pronged claw. A circular base provides strong support for the rest of the arm without restricting any degrees of freedom. The lower and upper limbs, having flexible joints, also allows the manipulator to engage with a wider range of motion. The manipulator's claw mechanism provides a variety of functions, such as grabbing and pushing. Our final design achieves 6 degrees of freedom using motors and hinges for the rotational axes.

The chosen material, 6061 aluminum alloy, is a material commonly used for milling parts. It is a durable and reliable material while still being lightweight. For our design, aluminum would be too heavy for our motor to correctly function. In the future, we would advocate for PET instead because it is 3D printable and lighter than aluminum. PET also meets our goal of being a smaller-scale arm for demonstrations. Another option is to opt for stronger motors that provide more torque to move the heavier materials.

Fabrication was tedious because coordinating a group of students throughout an asynchronous class is difficult. Sub-assemblies were assigned to each group member. Each member had to coordinate with other members to make sure the joints and assembly dimensions were accurate. Assembling the final model was also tedious as parts stopped working and had to be remade and reassembled. Despite the challenges of an asynchronous course, we managed as a group to complete this project and present the skills we learned in this course. We are proud of our final design and had a meaningful experience working on this project.

The robotic manipulator successfully achieved 6 degrees of freedom. Our maximum supported payload

is 46 grams. Although the supported weight isn't a lot, we learned that our design still has much room for improvement. There are many smaller details that we may have missed in our design process that most likely could have enhanced our arm's strength.

A major lesson we learned was that we should have done more research and design before we started our SolidWorks modeling. Diving straight into 3D modeling is more time-consuming, as you have to make adjustments to dimensions and parts each time something doesn't fit right or breaks. Whereas having a solid sketch and idea of what you want beforehand streamlines the 3D modeling process a lot. 3D modeling without initial agreements on dimensions proved to make our lives a lot harder during the assembly process. Early preparation is a good skill to be learned and applied.

Another lesson learned was the importance of good communication. Although this class was asynchronous and during the summer, this was still a group project, and we should have had better communication between group members. Everyone also has different standards of good work, and these should always be communicated beforehand. Lack of coordination increases stress and workload as a whole, not just for a singular person. We found ourselves to be more coordinated as the deadline rolled around, and that increase in communication benefitted our productivity a bunch.

Some takeaways for the future are that it is important to communicate well within your team and never be scared to overshare because every little detail counts. It is important to decide your material before you start your design process and make sure of the strengths and weaknesses of your motor to optimize payload. Most of the design work is done before you start 3D modeling as you plan out the goals and constraints of your project. Our group learned a lot about SolidWorks CAD and the engineering process through this project and is satisfied with our final design.

9. References

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10. Appendix

A. Parts Drawings:

In this appendix we post the engineering drawings of all the parts.

B. ASSEMBLY Drawing:

In this appendix we present the assembly drawings of the device.

11. Group Contributions

CAD - Subassemblies, Parts, Drawings

Base: Andrew Shentu
Lower Limb: Andrew Shentu
Upper Limb/Elbow Markus Lu
Claw: Aaron Wu

Report

Abstract - Aaron Wu
List of Figures - Markus Lu
List of Tables - Markus Lu
Introduction and General Background - Markus Lu
Prior Work - Aaron Wu
Design Concepts - Andrew Shentu, Aaron Wu
Theoretical Calculations - Markus Lu
Design Specs - Aaron Wu, Andrew Shentu, Markus Lu
Manufacturing and Fabrication- Andrew Shentu
Conclusion - Aaron Wu
Appendix - Andrew Shentu

Presentation

Aaron Wu, Andrew Shentu, Markus Lu